CRR Status Update

The Fourier Transform and the Role of ϕ

Alexander Sabine · Active Inference Institute · temporalgrammar.ai

Abstract

This note records two structural results regarding the Coherence-Rupture-Regeneration framework, identified through a careful comparison with classical Fourier analysis. First, the Fourier transform appears as the trivial CRR with linear coherence accumulation, imaginary Ω, and the causal arrow removed; CRR is therefore a strict generalisation of Fourier rather than an analogue. Twenty-five Fourier principles are reviewed, and CRR is found to generalise, reinterpret, or recover each. The deepest connection identified is with quasi-crystal spectral theory: ϕ-rotated CRR systems produce singular continuous spectra that match the empirical spectra of biological signals. Second, the long-standing notational coincidence between the resource field ϕ(x,τ) and the golden ratio ϕ ≈ 1.618 resolves into a single mathematical object: the eigenvector and eigenvalue, respectively, of the regeneration operator at minimal Markov-blanket depth. The pervasiveness of ϕ in nature is proposed as the pervasiveness of depth-two saturated open-system dynamics. Both results are presented as candidate structural claims, consistent with the framework as it stands and worth submitting to peer review.

1. Background and notation

The CRR framework is defined by three equations, reproduced here for reference.

$$C(x, t) = \int_0^t L(x, \tau)\, d\tau$$

(1) Coherence accumulation

δ(now), the Dirac delta encoding the instantaneous transition at C·Ω = 1

(2) Rupture

$$R[\chi](x, t) = \int_{-\infty}^{t} \phi(x, \tau) \cdot \exp(C(x, \tau)/\Omega) \cdot \Theta(t-\tau)\, d\tau$$

(3) Regeneration

Two CRR symmetry classes recur. Z_2 systems (bistable) have geodesic extent π, giving Ω = 1/π and rupture at C* = π. SO(2) systems (rotational) have geodesic extent 2π, giving Ω = 1/(2π) and rupture at C* = 2π. The ratio between classes is exactly two, a topological invariant. The beauty function B(C) = exp(C/Ω) · (C* − C) peaks at C* − Ω, one capacity-unit before rupture. The CV prediction CV = Ω/2 holds without free parameters.

2. The Fourier transform is the trivial CRR

2.1 The substitution

The Fourier transform takes the canonical form

$$F(\omega) = \int f(t) \cdot \exp(-i\omega t)\, dt$$

(4) Fourier transform

Comparison with the regeneration operator (3) reveals three structural differences: the Fourier kernel is complex-valued, whereas the CRR kernel is real-valued; the Fourier kernel uses a universal sinusoidal basis, whereas the CRR kernel is signal-specific through C(τ); and Fourier has no causal arrow, whereas

CRR has the Heaviside Θ.

Substituting $C(\tau) = \tau$ (trivial linear accumulation), $\Omega = i/k$ (rotation into the imaginary axis), and removing Θ:

$$exp(C/\Omega) = exp(\tau / (i/k)) = exp(-ik\tau)$$

(5) The Fourier kernel as trivial CRR

The Fourier transform falls out of CRR as the special case where the system has no specific coherence dynamics, the variance Ω is rotated into the complex plane, and the causal arrow is removed. CRR is therefore a strict generalisation, recovering Fourier as a limiting case. The non-trivial action of CRR is in systems where $C(\tau)$ is not a linear function of time, where Ω is a real positive quantity, and where the Heaviside Θ enforces causality.

2.2 Cramér-Rao and Heisenberg-Gabor are the same theorem

The CRR rupture condition $C \cdot \Omega = 1$ saturates the Cramér-Rao bound from statistical estimation theory. The corresponding Fourier-domain bound, the Heisenberg-Gabor uncertainty relation, asserts that for any signal the product of its time-spread and its frequency-spread is bounded below by 1/2:

$$(\text{time-spread}) \times (\text{frequency-spread}) \geq 1/2$$

(6) Time-frequency uncertainty (Heisenberg-Gabor)

Both bounds are consequences of Cauchy-Schwarz applied to conjugate variables, derived in different notation. Both saturate when the wave packet is Gaussian. The δ(now) rupture in CRR is geometrically the same object as the centre of a minimum-uncertainty Gabor wavelet. This is mathematical identity under appropriate substitution, not analogy.

3. CRR analysis of Fourier mathematics

A systematic review of twenty-five canonical Fourier principles, with the corresponding CRR statement for each. Connections fall into three categories: strict generalisation, reinterpretation of an existing theorem, or genuine new structural identity.

Fourier principle	CRR analogue or status
Fourier series (periodic)	Generalised to ϕ-rotated cycles; harmonics replaced by KAM-stable irrational ladders
Fourier transform	Trivial CRR: $C = \tau$, Ω imaginary, Θ removed
Convolution theorem	Holds conditionally: requires shared Ω between systems
Heisenberg-Gabor uncertainty	Same theorem as Cramér-Rao saturation $C \cdot \Omega = 1$
Parseval-Plancherel theorem	Fails: the deficit IS the second-law arrow
Poisson summation	Golden-ratio version: Birkhoff ergodic mean at ϕ-spaced points
Sampling theorem (Nyquist)	CRR $n = 1$ (one sample per Ω cycle): more compressive
Short-time Fourier transform	Symmetric CRR: special case with non-causal window
Wavelet transform	Multi-Ω CRR with hierarchical scale decomposition
Fractional Fourier transform	CRR parametrises the rotation angle by C/Ω dynamically
Hausdorff-Young inequality	Fails at the rupture point: phase-transition signature
Stein-Tomas restriction	Reduces to Cencov uniqueness for the Fisher-Rao metric

Pontryagin duality	CRR symmetry classes ARE the dual structure
Non-abelian Fourier	Open: corresponds to non-commuting CRR cycles
Gibbs phenomenon	Absent in CRR for its own basis representation of δ events
Spherical harmonics	CRR on geometric manifolds with appropriate symmetry
Bochner positive-definite kernels	CRR kernel is causal (one-sided), not positive-definite
Discrete Fourier transform	Golden-grid version: Fibonacci-indexed CRR-DFT
Quantum Fourier (QFT)	CRR may classically generate the position-momentum bound
Khinchin-Wiener spectrum	Lorentzian peaks plus 1/f broadband = nested CRR signature
Littlewood-Paley theory	CRR scales correspond to Littlewood-Paley bands
Calderon-Zygmund operators	CRR-CZ operators with Ω-dependent constants
T(1) theorem (David-Journé)	Dynamical stability ↔ L^2-boundedness of R
Carleson convergence	Stronger via exponential weighting: candidate L^1 result
Quasi-crystal spectra	EXACT match: singular continuous spectra of biological signals

3.1 Five identities, not analogies

Five connections are mathematical identities under appropriate substitution rather than loose analogies. First, Cramér-Rao bound (CRR's source) and Heisenberg-Gabor bound (Fourier's source) are the same theorem, both consequences of Cauchy-Schwarz on conjugate variables. Second, the fractional Fourier transform parametrises a rotation in the time-frequency plane by an angle in [0, π/2]; CRR with C accumulating from 0 to Ω traces out exactly this rotation, with $C \cdot \Omega = 1$ corresponding to angle π/4. Third, the Stein-Tomas restriction theorem applied to the rupture manifold reduces to Cencov's uniqueness theorem for the Fisher-Rao metric. Fourth, Pontryagin duality on locally compact abelian groups gives Z_2 dual to Z_2, and SO(2) dual to SO(2): these are exactly the CRR symmetry classes, and the relation $Z_2 + Z_2 \rightarrow SO(2)$ is a Pontryagin decomposition. Fifth, Khinchin-Wiener power spectra for a CRR oscillator are Lorentzians with width 1/Ω, and nested CRR cycles produce the 1/f broadband structure observed in biological signals.

4. The quasi-crystal spectrum result

Real biological signals exhibit power spectra that are neither pure point (no clean integer harmonics) nor absolutely continuous (preferred frequencies do exist). They are singular continuous, in the spectral classification of measure theory. Singular continuous spectra characterise quasi-crystals such as Fibonacci chains and Penrose tilings, where the relevant literature includes Bellissard, Damanik, and Lenz on the spectral theory of aperiodic order.

A ϕ-rotated CRR system is a quasi-crystal in time. Successive cycles are shifted by 1/ϕ, an irrational rotation that satisfies the Diophantine condition required for KAM stability. The resulting spectrum is forced to be singular continuous: the signal is neither periodic (which would give pure point spectrum) nor pure noise (which would give absolutely continuous spectrum). This matches the empirical 1/f noise structure of heart rate variability, EEG, neural avalanches, and other long-recording biological data.

The prediction is precise and testable. Biological signals driven by depth-two CRR at saturation should exhibit a spectrum that is mathematically classifiable as singular continuous in the same class as the Fibonacci chain. The 1/f exponent of the broadband structure should be approximately one for depth-two systems and should deviate predictably for systems with deeper memory.

5. The role of ϕ in CRR

5.1 Resolving the notational coincidence

Two distinct mathematical objects have shared the symbol ϕ within the CRR formalism. The first is the resource field appearing in the regeneration integral:

$$\phi(x, \tau) : \textit{a position-and-time-dependent reconstruction resource field}$$

(7) The resource field interpretation

The second is the golden ratio constant used in the temporal rotation grammar of successive CRR cycles:

$$\phi = (1 + \sqrt{5}) / 2 \approx 1.6180339887$$

(8) The golden ratio constant

These are different mathematical objects: one is a field, the other is a constant. The shared notation has been a source of conflation. The result presented here is that the two are not independent. They are the eigenvector and the eigenvalue, respectively, of the regeneration operator at minimal Markov-blanket depth.

5.2 The depth-two argument

A CRR system that regenerates sustainably must satisfy a self-consistency equation across cycles. The simplest non-trivial self-consistency that produces ergodic, KAM-stable behaviour is depth-two: the resource field at cycle n+1 depends on the field at cycles n and n–1.

$$r_{n+1} = \alpha \cdot r_n + \beta \cdot r_{n-1}$$

(9) Depth-two regeneration recurrence

Depth-one ($r_{n+1} = \alpha \cdot r_n$) gives geometric sequences with pure periodicity in log scale: not ergodic, not quasi-crystalline. Depth-three or greater gives ergodicity at the cost of higher reconstruction resources. Depth-two is the minimum depth that produces KAM-stable ergodicity, and it is also the minimum depth that makes the system non-Markovian with respect to its current state. This minimum-depth requirement is what distinguishes a sustainable open system with memory from either a Markovian current-state system or a chaotic deeper-memory system.

5.3 The eigenvalue calculation

The recurrence (9) can be written as a linear map on the state vector (r_n, r_{n-1}):

$$(r_{n+1}, r_n)^T = M \cdot (r_n, r_{n-1})^T,\ M = [[\alpha, \beta], [1, 0]]$$

(10) The companion matrix of the regeneration operator

The eigenvalues of M are the roots of the characteristic polynomial $\lambda^2 - \alpha\lambda - \beta = 0$:

$$\lambda = (\alpha \pm \sqrt{(\alpha^2 + 4\beta)}) / 2$$

(11) Eigenvalues of the depth-two regeneration operator

For the symmetric case $\alpha = \beta = 1$ (the simplest non-trivial regeneration: today's resources equal yesterday's plus the day-before's), the eigenvalues are:

$$\lambda_+ = (1 + \sqrt{5}) / 2 = \phi,\ \lambda_- = (1 - \sqrt{5}) / 2 = -1/\phi$$

(12) Eigenvalues at symmetric depth-two

The dominant eigenvalue is the golden ratio. The eigenvector corresponding to λ_+ is the resource field whose long-time dynamics is governed by ϕ. This is the field interpretation of $\phi(x, \tau)$. The two readings of the symbol are unified: the field is the eigenvector, the constant is the eigenvalue, and the regeneration operator is the linear map that connects them.

5.4 Why the dominant eigenvalue is ϕ specifically

The golden ratio satisfies the irreducible self-consistency equation

$$\phi^2 = \phi + 1, \text{ equivalently } 1/\phi = \phi - 1$$

(13) The defining equation of ϕ

By the Hurwitz theorem, ϕ has irrationality measure exactly one: it is the infimum over all irrationals of the constant c such that $|x - p/q| > c/q^2$ has solutions for infinitely many rationals p/q. In CRR terms, a system rotating at $1/\phi$ samples its own state space most uniformly: the orbit never returns close to where it was, always exploring new territory. This is what ergodicity means quantitatively, and ϕ-rotation is the most ergodic rotation possible.

5.5 What this implies

Three implications follow. First, the appearance of ϕ in the CRR temporal grammar is not arbitrary: it is forced as the dominant eigenvalue of the simplest sustainable regeneration operator. Second, the pervasiveness of golden-ratio scaling in nature, from phyllotaxis to vascular branching to neural fractal structure, is a consequence of biological systems operating at the minimum Markov-blanket depth that supports sustainable ergodicity. Third, deeper-memory systems should exhibit different growth constants. Depth-three with symmetric coefficients gives the tribonacci constant ≈ 1.83929 rather than ϕ. The 1/f exponent of a biological signal therefore predicts the effective memory depth of the underlying system.

6. Integration with the fine structure result

The fine structure constant derivation (Sabine, 2026) employs the same logical structure in a different physical regime. The QED electron-photon vertex is treated as a CRR cycle with vertex coherence

$$C_{vertex} = 2\pi^2 \alpha$$

(14) QED vertex coherence

where $2\pi^2$ arises from π (Z_2 vertex topology) times 2π (SO(2) loop integration), matching the CRR symmetry classes. Vacuum self-energy gives the dressed coherence $C_{dressed} = 2\pi^2 \alpha / (1 + (\pi-1)\alpha)$, with the coefficient $\pi-1$ fixed by the saddle point of the beauty function. The phase-space normalisation $16\pi^2 = (2\pi)^2 \times 4$ follows from angular and spin integration. The self-consistency requirement yields:

$$\alpha = \exp(2\pi^2 \alpha / (1 + (\pi-1)\alpha)) / (16\pi^2)$$

(15) The fine structure self-consistency equation

The unique stable fixed point of this equation is $1/\alpha = 137.032$, against the empirical value 137.036. The structure is identical to the eigenvalue argument for ϕ in Section 5: in both cases, a CRR system at saturation forces a specific real number as the unique stable solution of a self-consistency equation built from CRR axioms with no free parameters. ϕ is the eigenvalue for depth-two minimal regeneration; $1/\alpha$ is the fixed point for the QED vertex with vacuum self-energy. The general principle is that bounded self-consistent dynamics produce forced critical values.

7. Updated CRR status

7.1 What this update establishes

The Fourier review demonstrates that CRR connects substantively to twenty-five canonical principles of harmonic analysis. The connections fall into three categories: strict generalisations where CRR contains Fourier as a limiting case, reinterpretations where CRR offers a different proof or framing of an established theorem, and structural identities where two apparently independent results in different fields turn out to be the same theorem. The quasi-crystal spectrum match is the strongest of these connections, because it makes a precise prediction about real biological data that is testable using existing time-series and spectral analysis methods.

7.2 What this update resolves

The notational coincidence between the resource field $\phi(x,\tau)$ and the golden ratio constant $\phi \approx 1.618$ is resolved. The two are the eigenvector and eigenvalue of the regeneration operator at minimal Markov-blanket depth. The depth-two structure is the minimum that produces sustainable ergodicity, and the symmetric depth-two recurrence forces ϕ as the dominant eigenvalue. The pervasiveness of ϕ in biological scaling laws becomes the pervasiveness of minimum-depth saturated open-system dynamics.

7.3 What this update suggests for future work

Three lines of investigation are indicated. First, the quasi-crystal spectrum prediction should be tested against existing biological time-series databases: heart rate variability, long EEG recordings, and neural avalanche data. The signature of a singular continuous spectrum is technically distinguishable from pure 1/f noise and would provide independent confirmation of ϕ-rotated CRR. Second, the depth-prediction relationship suggests that different biological systems should exhibit different effective memory depths, with corresponding deviations from the ϕ scaling. The tribonacci constant (depth-three) and tetranacci constant (depth-four) are candidate eigenvalues for deeper-memory systems. Third, the integration with the fine structure result indicates a general programme: other fundamental constants may emerge as fixed points of CRR self-consistency equations with different symmetry classes. The programme is not a claim that this will succeed; it is a claim that the question is now well-posed and worth attempting.

7.4 Honest scope

CRR is a candidate temporal grammar for sustainable open systems with memory at saturation. It does not propose to be a theory of everything. It does not predict cosmological initial conditions, the values of arbitrary Standard Model parameters, the substrate of dark matter or dark energy, or the existence of phenomenal consciousness. It proposes that systems satisfying four conditions, namely openness to environment, bounded Markov blanket, memory depth at least two, and operation at the Cramér-Rao saturation surface, are governed by the structural relations described in equations (1) through (3), with consequences including the rupture condition $C \cdot \Omega = 1$, the beauty-function peak at $C^* - \Omega$, the parameter-free $CV = \Omega/2$, the symmetry-class ratio of two between Z_2 and $SO(2)$, and the ϕ-rotation grammar derived in Section 5. The Fourier connections in Section 3 and the fine structure connection in Section 6 are presented as further structural support, not as final proofs. The framework remains a candidate. The next step is peer review.

References

Bellissard, J., Iochum, B., Scoppola, E. & Testard, D. (1989). Spectral properties of one-dimensional quasi-crystals. *Comm. Math. Phys.*, 125, 527-543.

Cencov, N. N. (1982). *Statistical Decision Rules and Optimal Inference*. American Mathematical Society.

Damanik, D. (2007). Strictly ergodic subshifts and associated operators. *Spectral Theory and Mathematical Physics*, 76, 505-538.

Friston, K. (2010). The free-energy principle. *Nature Reviews Neuroscience*, 11, 127-138.

Hurwitz, A. (1891). Ueber die angenäherte Darstellung der Irrationalzahlen durch rationale Brüche. *Math. Ann.*, 39, 279-284.

Ito, S. & Dechant, A. (2020). Stochastic time evolution and the Cramér-Rao bound. *Phys. Rev. X*, 10, 021056.

Lenz, D. (2002). Singular continuous spectrum of Lebesgue measure zero for one-dimensional quasi-crystals. *Comm. Math. Phys.*, 227, 119-130.

Parr, T., Pezzulo, G. & Friston, K. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.

Sabine, A. (2026). The fine structure constant from CRR self-consistency. temporalgrammar.ai.

Sabine, A. (2026). Phase-gating across precision channels. AGI-26 submission.

Whitehead, A. N. (1929). *Process and Reality*. Macmillan.